



Faculty of Computer Science

Master of Data Science

Applied Quantitative Logistics

Genetic Speciation Algorithm for Addressing Combinatorial Optimization Problems: Application to the Vehicle Routing Problem



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Problem Definition

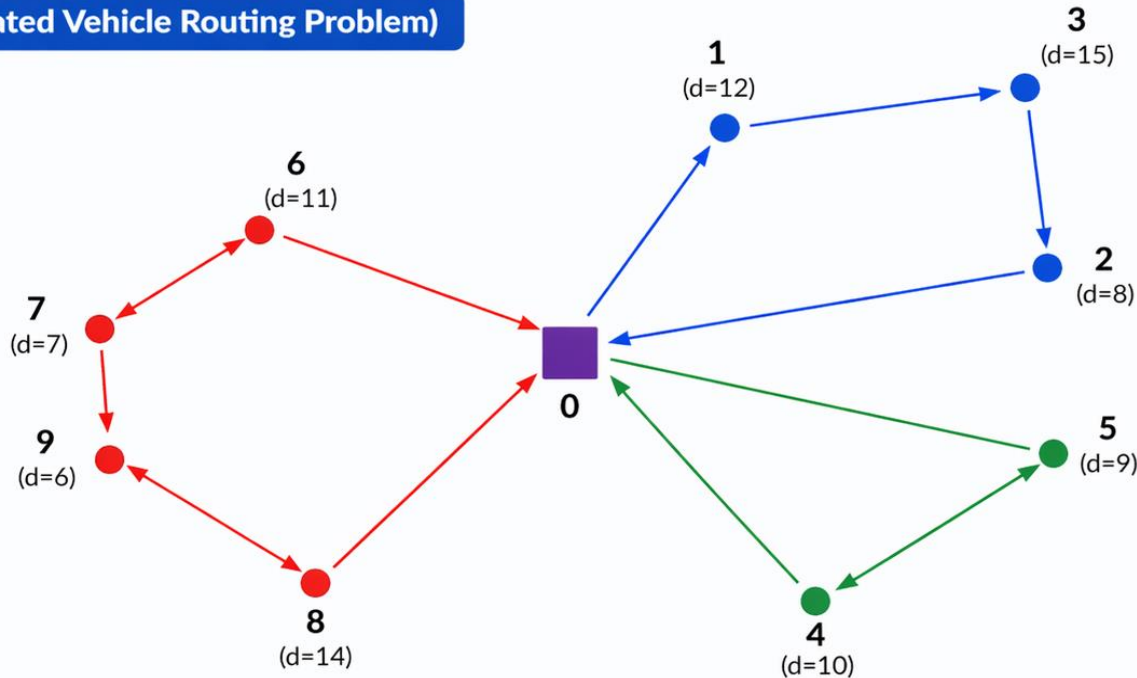
What is CVRP?

- A fleet of vehicles departs from a **single depot** to serve a **set of customers**
- Each customer has a **fixed demand**; each vehicle has a **capacity limit**
- Objective: **minimize total travel distance** across all routes

Why It is Hard?

- **NP-hard** combinatorial optimization problem
- **Search space grows factorially** with customer count

CVRP (Capacitated Vehicle Routing Problem)



Route 1

0 → 1 → 3 → 2 → 0 Distance = 31

Route 2

0 → 5 → 4 → 0 Distance = 18

Route 3

0 → 6 → 7 → 9 → 8 → 0 Distance = 29

Total Distance = 31 + 18 + 29 = 78

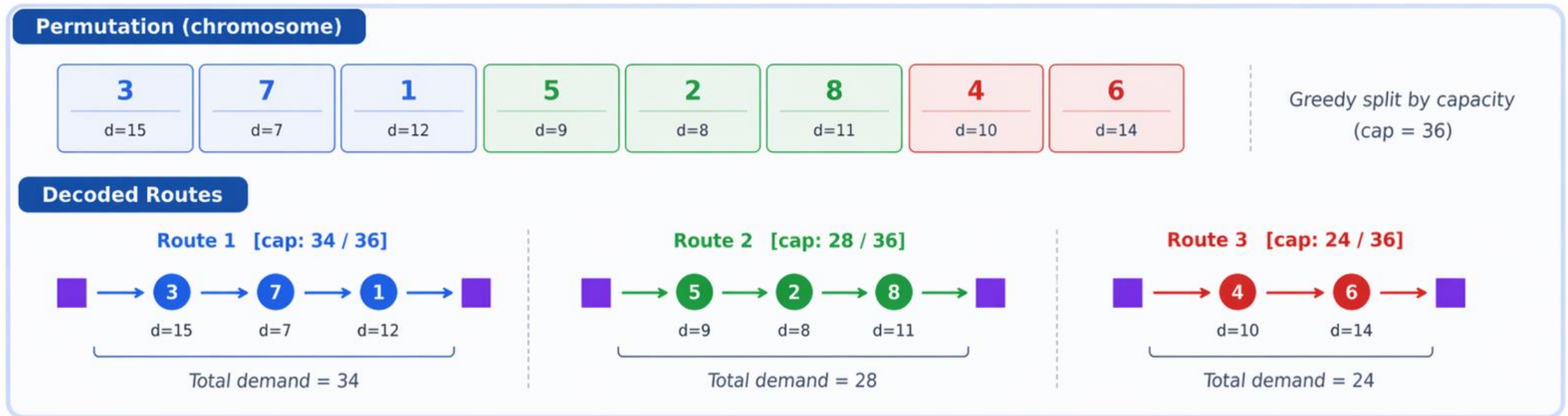
Solution Representation

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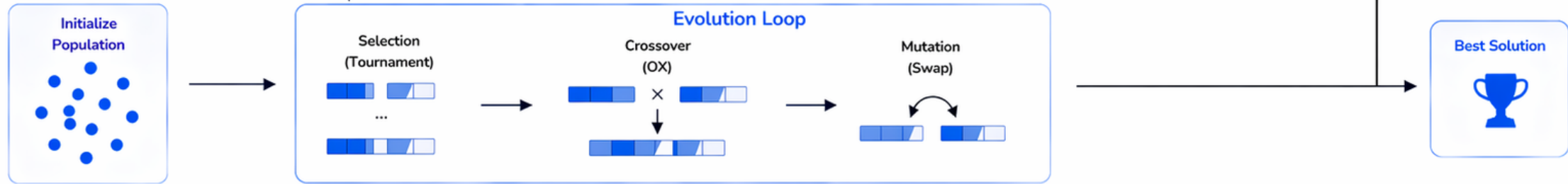
- A solution is encoded as a **permutation of all customers**
- Routes are **decoded greedily**: a new route begins whenever adding the next customer would exceed **vehicle capacity**
- **Fitness = total route length including depot returns**

Fitness Evaluation

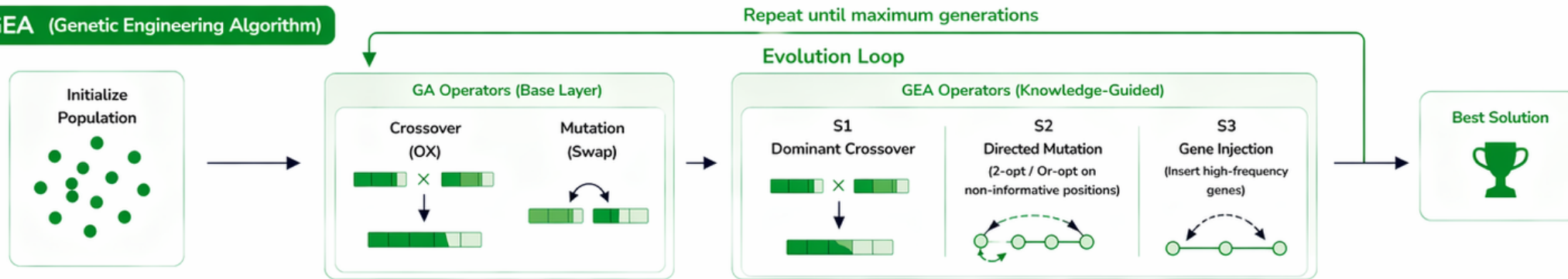
- **Single-pass** over the permutation — **no $O(n^2)$ feasibility checks**
- **Most frequently called function** in the entire pipeline; kept **minimal for performance**



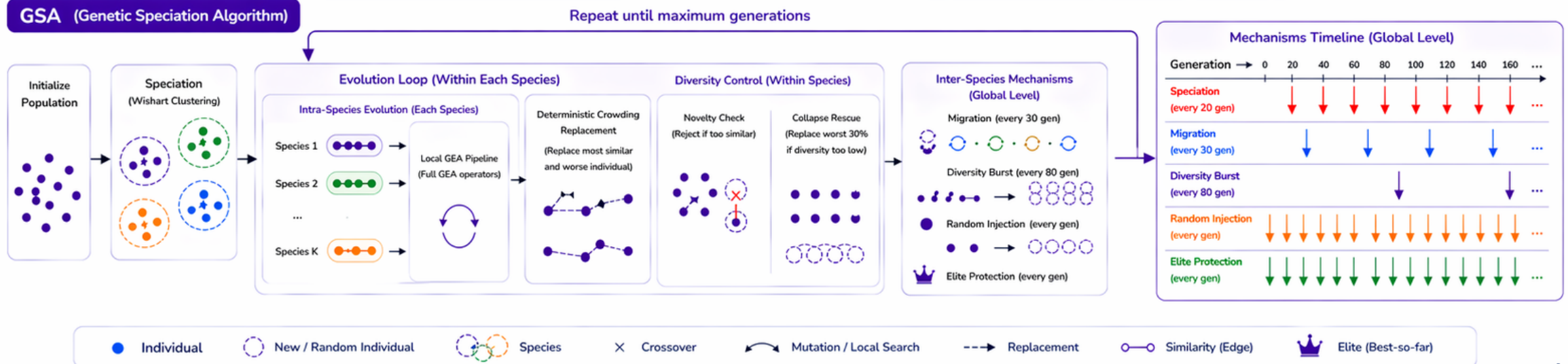
GA (Genetic Algorithm)



GEA (Genetic Engineering Algorithm)



GSA (Genetic Speciation Algorithm)



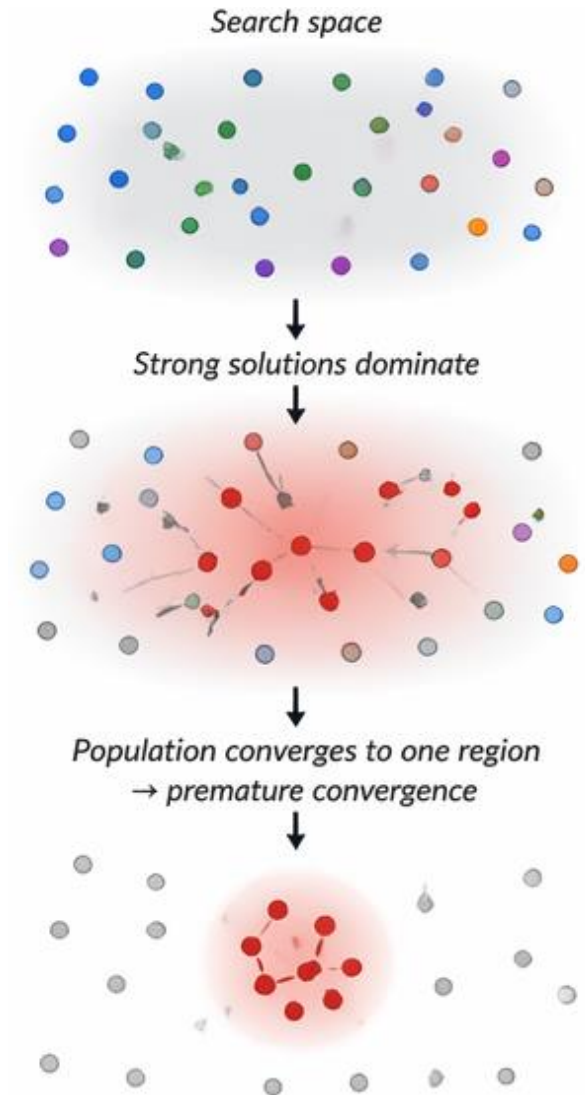
Wishart Speciation

Why Speciation?

- **Single-population methods** force all individuals to compete for the same slots
- **Strong solutions dominate** → **structurally different individuals are eliminated**
- The population **converges to one region** of the search space → **premature convergence**
- **Speciation enforces structural separation**: diverse archetypes coexist and evolve in parallel

How to measure similarity between individuals?

- Primary (80%): **Jaccard distance between edge sets** — two individuals are close if they share many consecutive customer pairs
 - **Jaccard Similarity**: $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$
 - **Jaccard Distance**: $D(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}$
- Secondary (20%): **fitness rank distance** — prevents grouping individuals of very different quality together
- **Combined**: $D = 0.8 \times \text{Jaccard} + 0.2 \times \text{rank distance}$



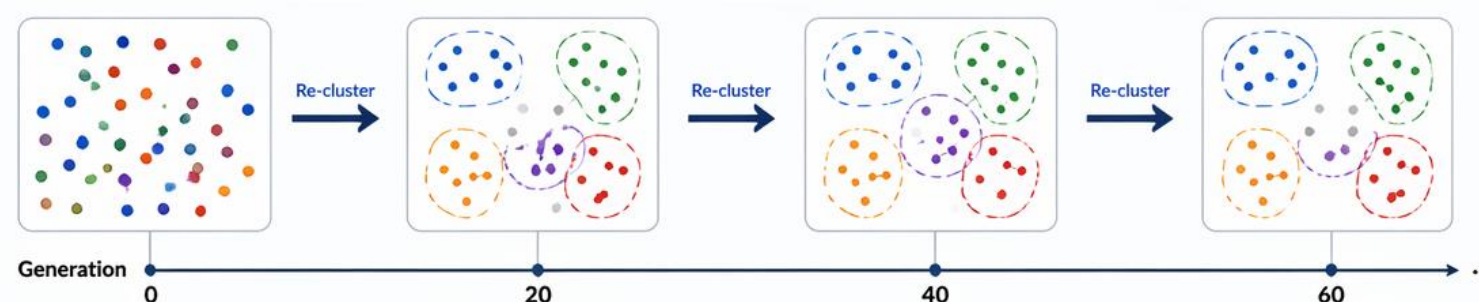
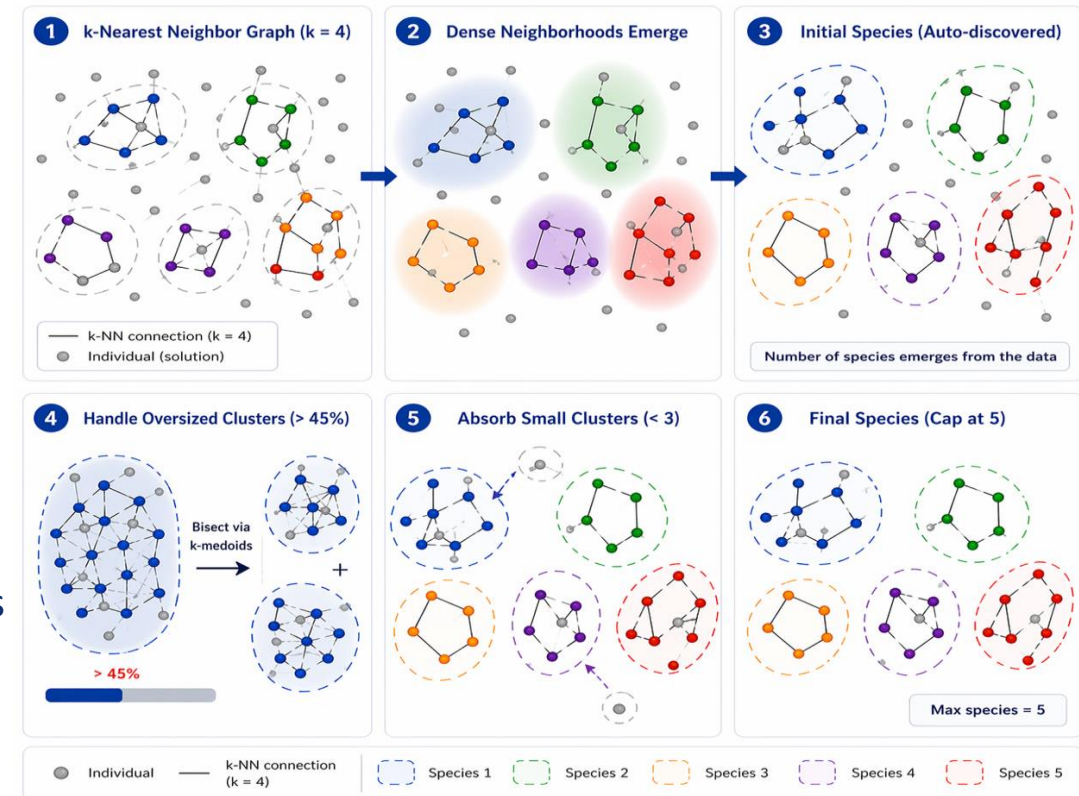
Wishart Speciation

How species are formed?

- Connect each individual to its **k nearest neighbors** ($k = 4$)
- Clusters form around **dense neighborhoods** and **expand naturally**
- Number of species **emerges from the data** — not fixed in advance
- **Oversized clusters** ($> 45\%$ of population) are bisected via k-medoids
- **Clusters smaller than 3 individuals** are absorbed into the nearest species
- Maximum species count capped at 5 to **prevent over-fragmentation**

When to Re-cluster?

- **Every 20 generations:** routine re-clustering to reflect **evolved population structure**
- **Immediately after a diversity burst:** population has **changed drastically**, old species labels are invalid



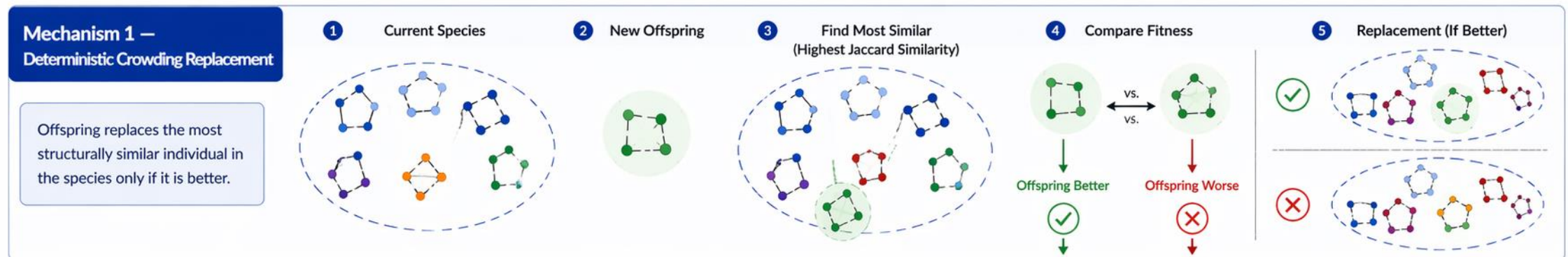
Intra-Species Diversity Control

Why Not Just Replace the Worst?

- GEA uses **replace-worst**: an offspring replaces the globally worst individual if it is better
- This **works in a single population** but breaks down inside a species — the worst individual may be structurally very different from the offspring, so replacing it **destroys diversity** rather than improving it
- GSA uses **three targeted mechanisms** instead

Mechanism 1 — Deterministic Crowding Replacement

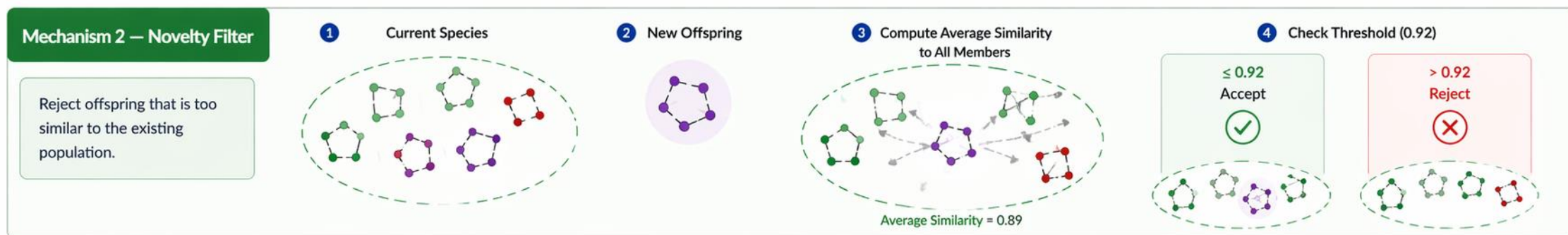
- When an offspring is produced, **find the most structurally similar individual** in the species (highest Jaccard similarity)
- The offspring replaces that individual **only if it has better fitness**
- Effect: new solutions **displace their structural competitors**, not unrelated individuals — the species **retains its structural spread**



Intra-Species Diversity Control

Mechanism 2 — Novelty Filter

- Before accepting an offspring into the species, check its **average similarity** to all current members
- If the offspring is too similar to the existing population (**similarity > 0.92**), **reject** it outright
- Effect: prevents the species from **collapsing to a single structure** even under strong selective pressure



Mechanism 3 — Collapse Rescue

- Every generation, compute the **average pairwise Jaccard distance** within each species
- If this **falls below 0.04** (near-identical structures across all members), trigger a rescue
- Replace **the worst 30%** of the species with randomly generated individuals
- Effect: **breaks up clone clusters** before they waste further generations going nowhere

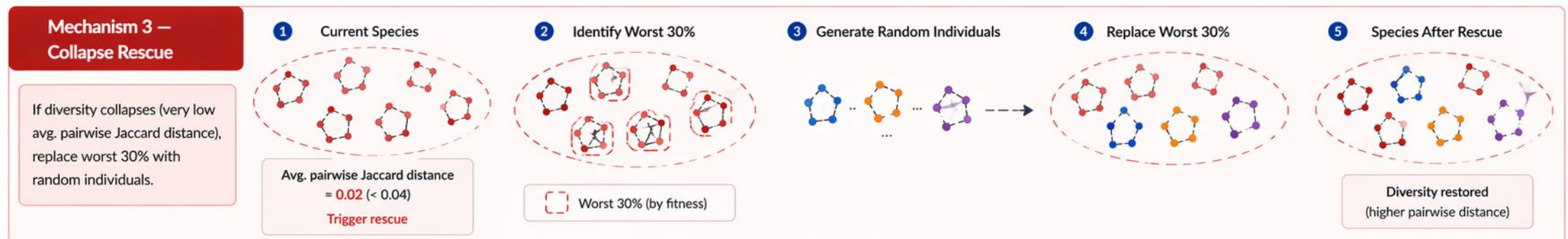
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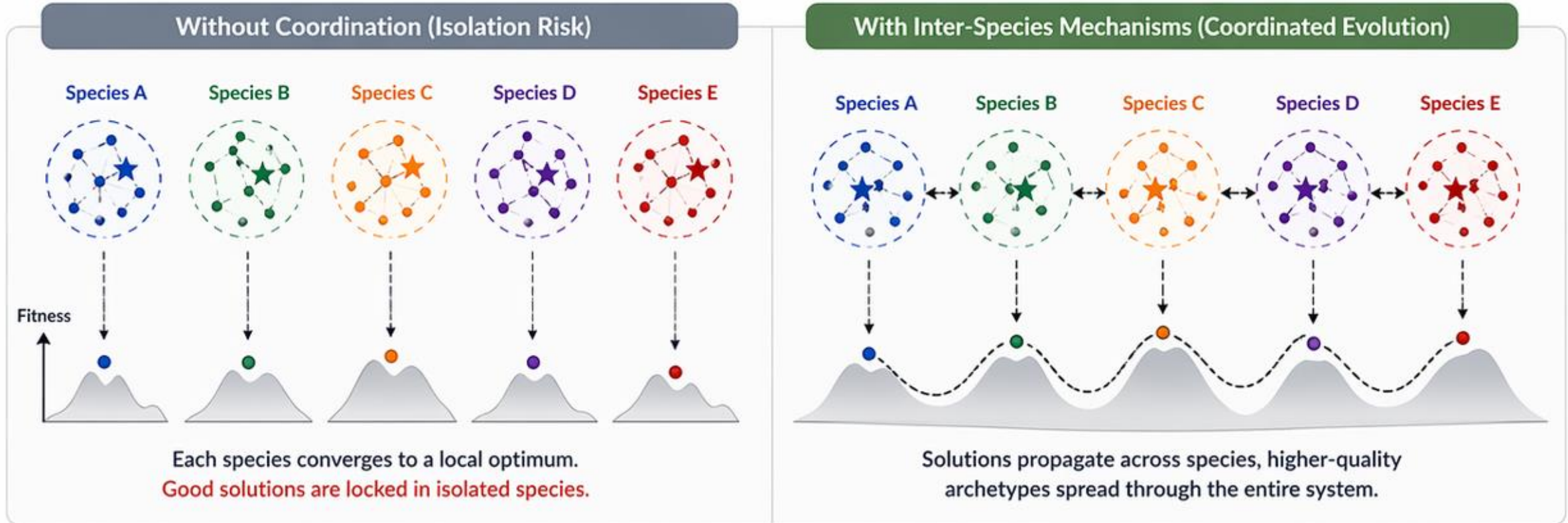
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Inter-Species Mechanisms

Why Inter-Species Mechanisms?

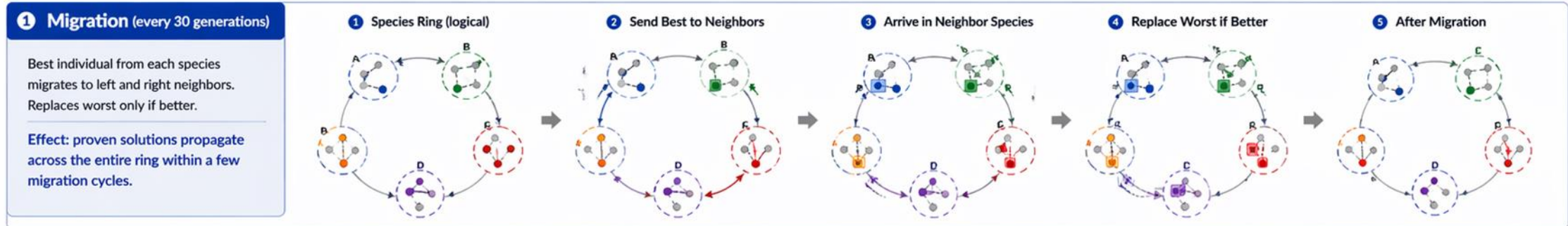
- **Independent species evolution risks isolation** — a species may converge to a **locally strong but globally inferior solution** archetype
- **Periodic cross-species mechanisms** enable **information sharing** and prevent **global stagnation**



Inter-Species Mechanisms

Migration (every 30 generations)

- Species are **arranged in a logical ring**
- The **best individual** from each species is sent to both its left and right neighbors
- In each receiving species, the migrant replaces the worst member **only if it has better fitness**
- Effect: proven solutions propagate across the entire ring within a few migration cycles — **good structural patterns are not locked inside a single species**



Diversity Burst (every 80 generations)

- The **bottom 25% of the global population** is replaced with **randomly generated individuals**
- Forces a re-clustering on the next cycle, introducing **entirely new structural material**
- Effect: **resets stagnating species** before they waste too many generations; the long interval (80 gens) avoids disrupting ongoing convergence

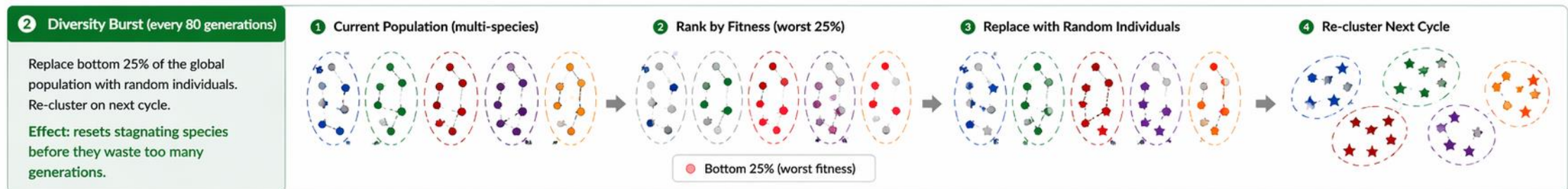
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Inter-Species Mechanisms

Random Injection (every generation)

- The **worst 3% of the global population** is replaced with **random individuals**, but only if the random individual is actually better
- A lightweight continuous mechanism providing a steady trickle of **new genetic material**
- Effect: **prevents full stagnation** without disturbing the overall population structure



Elite Protection (every generation)

- The **global best solution** found so far is stored at the start of each generation
- After all operators have run, if the global best has been **accidentally overwritten**, it is restored
- Effect: guarantees monotonic improvement in reported best fitness — **no good solution is ever permanently lost**



Experimental Setup

Benchmark Instances

Instance	Customers	Vehicles
A-n32-k5	31	5
A-n60-k9	59	9
A-n80-k10	79	10

Algorithms Compared

- **GA:** crossover + mutation only (baseline)
- **GEA:** GA + elite knowledge extraction + S1/S2/S3 operators
- **GSA:** GEA + Wishart speciation + intra-species diversity control + inter-species mechanisms

Evaluation Metrics

- Best: **best solution** found across all 5 runs
- Mean: **average solution quality** across runs
- Std: **standard deviation** across runs (consistency)
- Avg Time: **average wall-clock time** per run in seconds

Hyperparameters

Parameter	Value
Population size	200
Generations	500
Independent runs	5
Base random seed	42
Elite ratio	20% (GEA) / 40% (GSA)
Max species	5
Re-cluster every	20 generations

Reproducibility

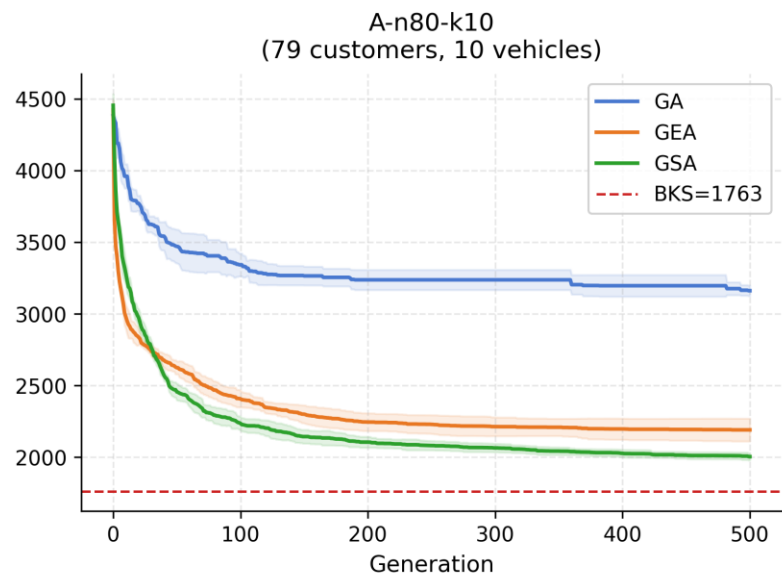
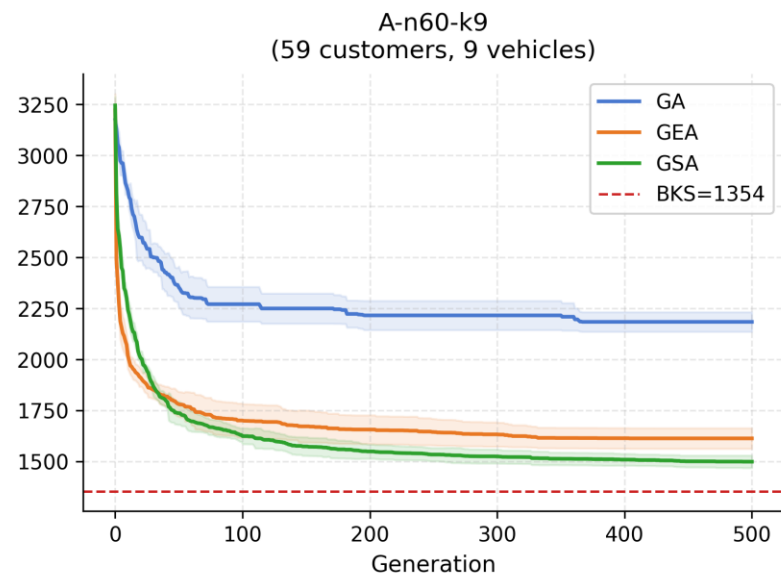
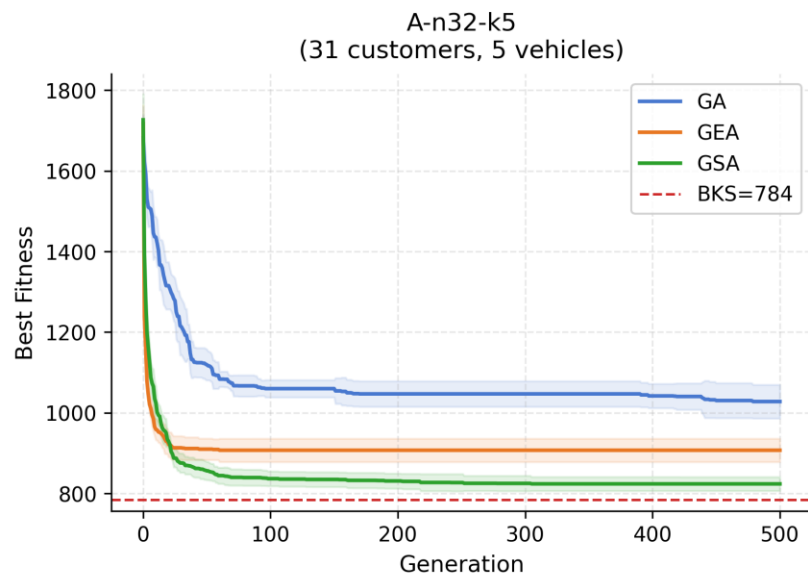
- Each algorithm and run uses a distinct **random seed** derived from the base seed, instance name, and run index
- Every reported solution is validated before recording:
 - permutation must be **complete**
 - every route must **satisfy capacity constraints**



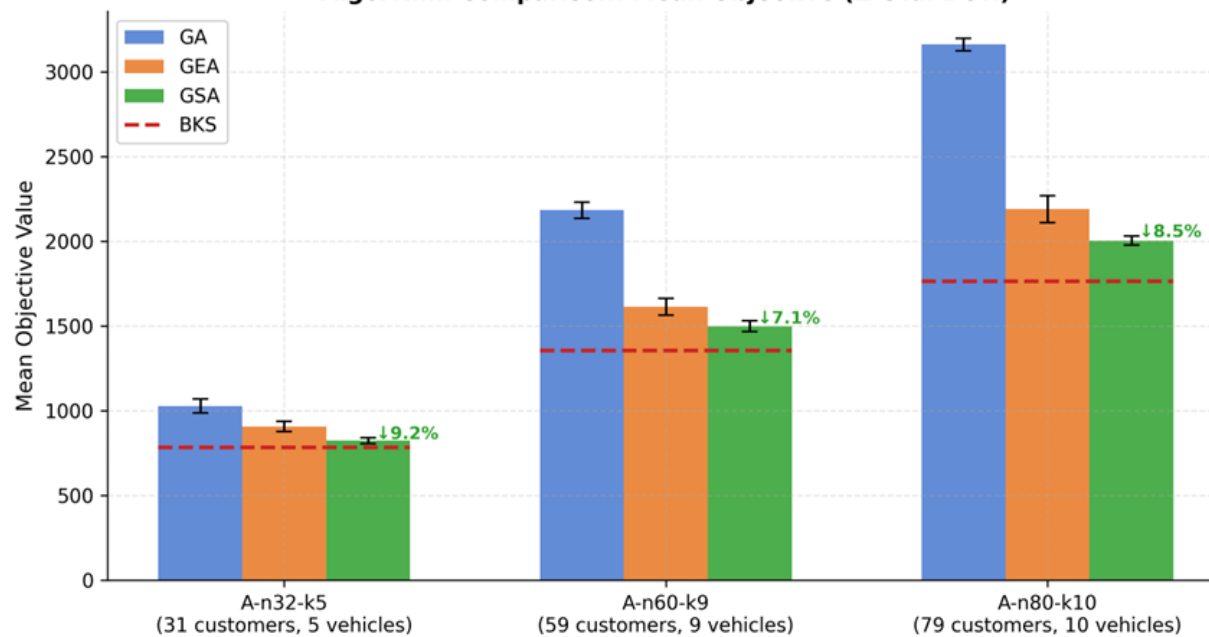
Experimental Results

Instance	Algorithm	Best	Mean	Std. Dev.	Gap to BKS	Avg. Time
A-n32-k5	GA	944.6	1027.9	41.79	31.1%	4.5s
A-n32-k5	GEA	857.6	907.1	28.99	15.7%	12.1s
A-n32-k5	GSA	803.3	823.6	17.27	5.1%	55.5s
A-n60-k9	GA	2110.1	2184.1	48.12	61.3%	6.8s
A-n60-k9	GEA	1517.2	1613.0	50.63	19.1%	17.5s
A-n60-k9	GSA	1463.7	1499.0	30.72	10.7%	90.9s
A-n80-k10	GA	3104.7	3161.3	36.98	79.3%	8.0s
A-n80-k10	GEA	2078.1	2190.9	79.58	24.3%	20.7s
A-n80-k10	GSA	1978.1	2004.2	25.30	13.7%	132.4s

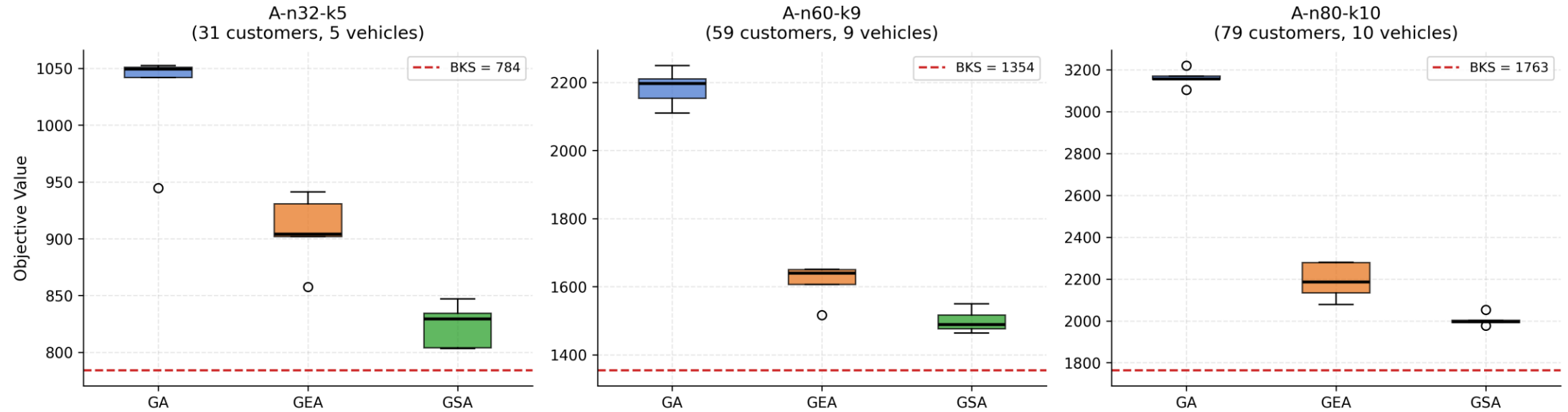
Mean Convergence Curves (± 1 Std. Dev.)



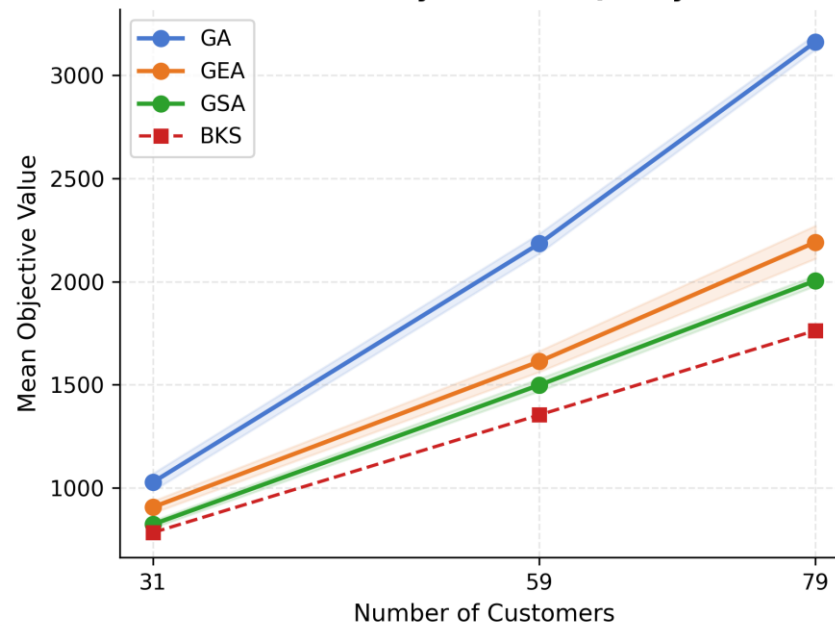
Algorithm Comparison: Mean Objective ($\pm$ Std. Dev.)



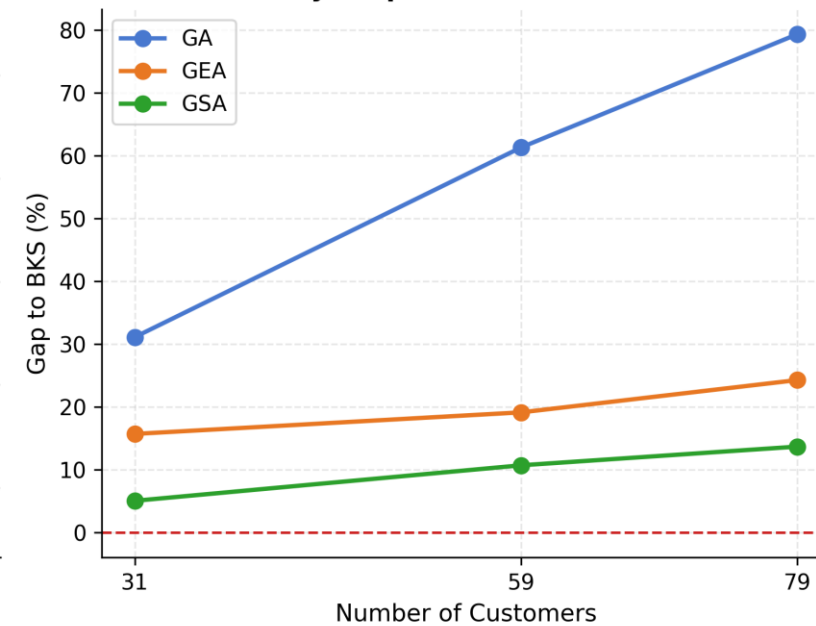
Result Distribution Across Independent Runs



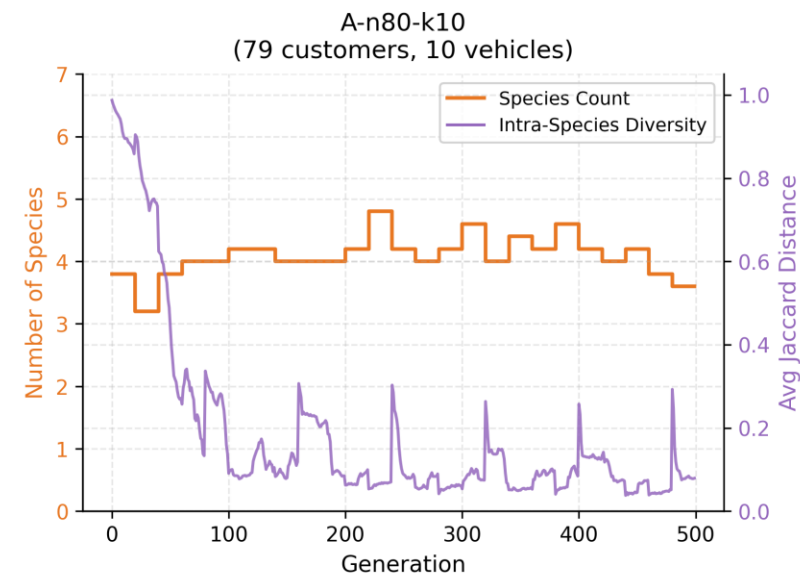
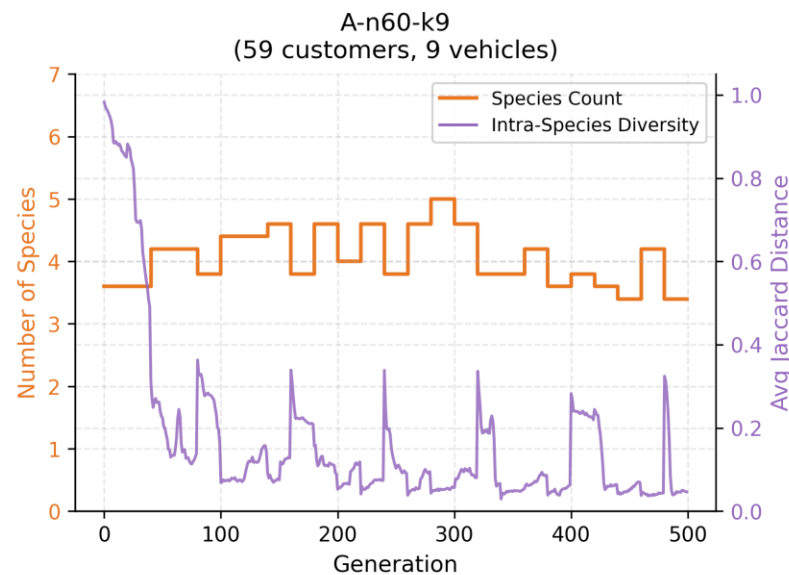
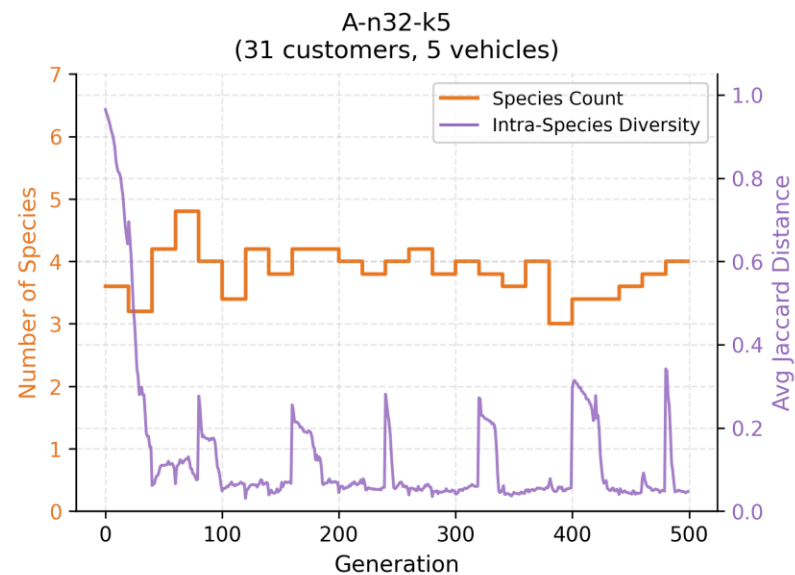
Scalability: Solution Quality



Scalability: Gap to Best Known Solution



GSA Species Dynamics (Averaged over All Runs)



GSA Species Size Evolution (Run 1)

